

Paper: 1MA1/3H					
Question		Answer	Mark	Mark scheme	Additional guidance
16	(a)	129 to 160	M1	for a method to find an estimate for the area under the curve eg $0.5 \times 30 \times 9$	Do not accept 30×9 Award full marks for any correct method leading to a better estimate.
			A1	for value in the range 129 to 160 (If M0, SC B1 for 126 or 127.5)	
	(b)	underestimate with reason	C1	(dep M1) for “underestimate” and appropriate reason linked to their method, eg area between triangle and curve not included	
	(c)	Explanation	C1	for explanation, Acceptable examples method gives average acceleration (in first 60 seconds) he has not used/drawn a tangent (at time 60 seconds) he has not worked out the gradient (at time 60 seconds) Not acceptable examples he has not used strips he has calculated it accurately rather than using an estimate the estimate of 13 should be about 4.4 the answer should be approximately 0.073	